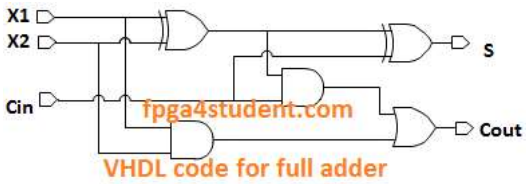
VHDL code for Full Adder

In this VHDL project, VHDL code for full adder is presented. VHDL code for the adder is implemented by using behavioral and structural models.

The full adder has three inputs X1, X2, Carry-In Cin and two outputs S, Carry-Out Cout as shown in the following figure:



The VHDL code for the full adder using the structural model:

```
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-- VHDL code for full adder
-- Structural code for full adder
 library ieee;
 use ieee.std_logic_1164.all;
 entity Full_Adder_Structural_VHDL is
   port(
  X1, X2, Cin : in std_logic;
  S, Cout : out std_logic
  );
 end Full_Adder_Structural_VHDL;
 architecture structural of Full_Adder_Structural_VHDL is
 signal a1, a2, a3: std_logic;
 begin
   a1 <= X1 xor X2;
   a2 <= X1 and X2;
   a3 <= a1 and Cin;
   Cout <= a2 or a3;
   S <= a1 xor Cin;
 end structural;
Library IEEE;
USE IEEE.Std_logic_1164.all;
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-- VHDL code for full adder
-- Testbench code of the structural code for full adder
entity Testbench_structural_adder is
end Testbench_structural_adder;

architecture  behavioral of Testbench structural_adder is
 component Full_Adder_Structural_VHDL
   port(
  X1, X2, Cin : in std_logic;
  S, Cout : out std_logic
  );
 end component;
 signal A,B,Cin: std_logic:='0';
 signal S,Cout: std_logic;
begin
 structural_adder: Full_Adder_Structural_VHDL port map
   (
    X1 => A,
    X2 => B,
    Cin => Cin,
    S => S,
    Cout => Cout
   );
  process
  begin
   A <= '0';
   B <= '0';
```

```
    Cin <= '0';
    wait for 100 ns;
    A <= '0';
    B <= '0';
    Cin <= '1';
    wait for 100 ns;
    A <= '0';
    B <= '1';
    Cin <= '0';
    wait for 100 ns;
    A <= '0';
    B <= '1';
    Cin <= '1';
    wait for 100 ns;
    A <= '1';
    B <= '0';
    Cin <= '0';
    wait for 100 ns;
    A <= '1';
    B <= '0';
    Cin <= '1';
    wait for 100 ns;
    A <= '1';
    B <= '1';
    Cin <= '0';
    wait for 100 ns;
    A <= '1';
    B <= '1';
    Cin <= '1';
    wait for 100 ns;
   end process;

 end behavioral;
```

Simulation waveform of the structural VHDL code for the full adder:

The VHDL code for the full adder using the behavioral model:

```
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 -- VHDL code for full adder
 -- Behavioral code for full adder
  library ieee;
  use ieee.std_logic_1164.all;
  use IEEE.STD_LOGIC_unsigned.ALL;
 use IEEE.NUMERIC_STD.ALL;
  entity Full_Adder_Behavioral_VHDL is
    port(
   X1, X2, Cin : in std_logic;
   S, Cout : out std_logic
   );
  end Full_Adder_Behavioral_VHDL;
  architecture Behavioral of Full_Adder_Behavioral_VHDL is
  signal tmp: std_logic_vector(1 downto 0);
  begin
    process(X1,X2,Cin)
    begin
  tmp <= ('0'& X1) + ('0'& X2) +('0'& Cin) ;
    end process;
    S <= tmp(0);
    Cout <= tmp(1);
  end Behavioral;
  Library IEEE;
 USE IEEE.Std_logic_1164.all;
```

```
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-- VHDL code for full adder
-- Testbench code of the behavioral code for full adder
entity Testbench_behavioral_adder is
end Testbench_behavioral_adder;

architecture  behavioral of Testbench_behavioral_adder is
 component Full_Adder_Behavioral_VHDL
   port(
  X1, X2, Cin : in std_logic;
  S, Cout : out std_logic
  );
 end component;
 signal A,B,Cin: std_logic:='0';
 signal S,Cout: std_logic;
begin
 behavior_adder: Full_Adder_Behavioral_VHDL port map
   (
    X1 => A,
    X2 => B,
    Cin => Cin,
    S => S,
    Cout => Cout
   );
  process
  begin
   A <= '1';
   B <= '1';
   Cin <= '1';
   wait for 50 ns;
   A <= '1';
   B <= '1';
   Cin <= '0';
   wait for 50 ns;
   A <= '1';
   B <= '0';
   Cin <= '1';
   wait for 50 ns;
   A <= '0';
   B <= '0';
   Cin <= '0';
   wait for 50 ns;
   A <= '0';
   B <= '0';
   Cin <= '1';
   wait for 50 ns;
   A <= '0';
   B <= '1';
   Cin <= '0';
   wait for 50 ns;
   A <= '0';
   B <= '1';
   Cin <= '1';
   wait for 50 ns;
   A <= '1';
   B <= '0';
   Cin <= '0';
   wait for 50 ns;

  end process;

end behavioral;
```
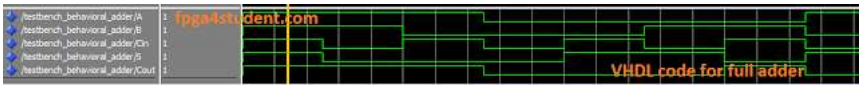

Simulation waveform of the behavioral VHDL code for the full adder:



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